Project Charter

Agile Software Engineering Capstone

Document field	Team entry
Project title	Student Savings Project
Team name	Ninja Penguins
Team members	Lupe Campos Joey Contreras Doga Eski Jonathan Ochoa
Project manager or team lead	Jonathan Ochoa
Project sponsor	Commerce Bank
Customer or client	Commerce Bank Banking Users
Instructor	Syed Jawad Hussain Shah
Start and target end dates	Start: August 24, 2026 End: December 7, 2026 (Last week of semester)
Version status and date	Complete 11/6/2026

Revision history

Version	Date	Author	Change and approval
0.1	9-25-26	Team	Initial draft
0.2	11-5-26	Lupe	Final review of all content.

1 Project Purpose and Authorization

This section explains why the project should exist and gives the team authority to proceed. Describe the problem before proposing detailed features.

1.1 Problem or opportunity

[In 3 to 5 sentences, explain who experiences the problem, what happens today, and why it matters. Do not describe the solution yet.]

People who manage personal finances can have trouble tracking where their money is being spent. When expenses such as groceries, transportation, loans, and other necessities are spread across savings, checking, and credit cards it can be difficult to have a clear view of which expenses are too much, or too little. This makes it difficult to manage monthly expenses and make informed decisions about money.

1.2 Project justification

Decision test	Team explanation	Evidence or source
Needed	Student budgeting	Student debt is all too common in the US. Our app is aimed at helping them better manage debt and finances. https://educationdata.org/average-student-loan-debt
Feasible	We can complete this product as we intend to focus on the basic financing that comes with being a student.	Using Java, Railway, and springboot we will complete this web application to help students navigate their finances.
Optimum	This project would enable a lot of students to easily manage their finances as a lot of students are unfamiliar with loans and budgeting fresh out of high school.	Our web application is a student friendly way to better manage finances with the help of an AI assistant and visual aids.

1.3 Product vision

Our app will help students budget for student loans and any other financial goals.

1.4 Authority to proceed

Authorization item	Entry
Decision	Approved to proceed
Conditions	The team must follow the approved project scope, complete requirements and project planning documentation, communicate progress during each sprint, and test features before the final demonstration.
Authority granted	The team is authorized to proceed with project development, database setup, user authentication, dashboard implementation, financial management features, and sprint planning.

2 Stakeholders and Team Authority

Identify everyone who can affect the project or is affected by its outcome. Record their needs and decision authority so the team knows whom to consult.

2.1 Stakeholder register

Stakeholder or group	Interest needs or expectations	Decision or contribution	Contact or review cadence
Project sponsor	Commerce Bank	Introduced need and does final check on project.	Becca Fantasma Anar Agayev Clayton Sloniker
Customer or client	Commerce Bank	Commerce Bank	Becca Fantasma Anar Agayev Clayton Sloniker
Target users	Budgeting app for college students	Provide feedback on UI	Towards the end of development of each sprint.
İnstructor	Syed Jawad Hussain Shah	Clarifies course scope and assesses work	Office hours or course channel
Student team	Refer to section 2.2 Team roles and responsibilities.	Refer to section 2.2 Team roles and responsibilities.	Microsoft Teams
[Other stakeholder]	[Needs and expectations]	[Decision or contribution]	[Contact or cadence]

2.2 Team roles and responsibilities

Team member	Primary role	Charter responsibility	Decision authority or backup
Jonathan	Project Manager	Maintains direction and scope of project	Main decision maker
Doga	Front-end development	Front end development of project	Final decision on UI of our app.
Joey	Back-end development	Development of back-end and anything managing our user's data.	Final decision on data management in our app.
Lupe	Documentation	Documenting anything related to the project.	Final decision on documentation and back up for the front-end development.

2.3 Decision and escalation rules

The team routinely works out details before moving on with anything and any disagreements will be up to the project manager. If it cannot be resolved by that, then we would go to the instructor.

3 Goals Objectives and High Level Features

Goals describe the desired direction. Objectives make the expected outcomes specific and measurable. Features identify the main functions that deliver value to the client or users.

3.1 Project goals

Goal ID	Goal statement	Why it matters
G-01	Develop web/app for students to use for managing their personal finances.	Provides students with an accessible way to organize their finances
G-02	Provide students with an easy-to-use platform for tracking spending, creating budgets, and managing loans.	Helps students monitor their financial activity and make informed decisions on their spending plans
G-03	Provide personalized financial assistance with the help of an AI-powered support module	Gives students additional guidance and makes financial info easier to understand

3.2 Measurable objectives

Objective ID	Specific measurable objective	Target date	How the team will verify it
OBJ-01	Implement registration and login functionality	Sprint 1: September 28 – October 2	Demonstration and testing of registration, login, logout, forgot password, and password reset procedure
OBJ-02	Complete the user authentication system and establish the database	Sprint 2: October 5 – October 9	Verify database connectivity and test login functionality and planned authentication features
OBJ-3	Implement the dashboard and forgot password functionality	Sprint 3: October 12 – October 16	Demonstration and testing of the dashboard, password recovery, and dashboard features

OBJ-4	Complete most of the user profile, settings, and additional dashboard functionality	Sprint 4: October 19 – October 23	Demonstration of profile and settings features, and backend functionality
OBJ-5	Complete loan, budget, goal, and account pages	Sprint 5: October 26 – October 30	Finalizing current features for demonstration, integration testing, and verification that all major features work as expected.

3.3 High level features

Feature ID	Client valued feature	Primary user or stakeholder	Priority
F-01	User registration, login, logout, and password recovery	Students/Users	Must
F-02	User profile and account settings management	Students/Users	Must
F-03	Monthly budget creation and category spending limits	Students/Users	Must
F-04	Dashboard displaying financial summaries and spending activity	Students/Users	Must
F-05	Financial account management	Students/Users	Must
F-06	Income and expense transaction management	Students/Users	Must
F-07	Loan management and payment tracking	Students/Users	Must
F-08	Financial goals creation and progress tracking	Students/Users	Must
F-09	AI-powered financial assistance and spending summaries	Students/Users	Must
F-10	Notifications and account alerts	Students/Users	Should

4 Scope Priorities and Deliverables

4.1 Scope boundaries

In scope	Out of scope
- A responsive web application for budgeting and personal finance tracking, built for college students	- Real banking connections, real money movement and real card data - Direct integration with Commerce Bank's live production or core banking systems

- User registration, login, logout and password recovery (F-01) - User profile and account settings management (F-02) - Loan management and payment tracking: balances, payment schedules, interest and payoff progress (F-03) - Budget creation and categorized expense tracking (groceries, transportation, loans, etc.) using sample transaction data(F-03) - Dashboard visualizing spending against budget by category(F-04) - AI-powered support module that gives students plain-language budgeting and loan guidance (F-09) - PostgreSQL database for storing user, budget and loan data - Core automated testing (unit and integration) for the features above	- Investment management, tax filing or credit score monitoring features - Native IOS/Android apps (web only for this semester) - Multi-currency or international banking support - Regulatory/Compliance certification and production-grade fraude detection

4.2 Project priorities

Rank the priorities from 1 to 3. Use each rank once, then explain the tradeoff rule.

Priority area	Rank	Team agreement
Scope and product value	3	Scope is the variable that flexes. Must-have features (F-01 through F-09) are protected; should have features (advanced AI guidance) are the first tobe trimmed if schedule or quality are at risk.
Schedule and course deadlines	1	The semester end date (December 11, 2026) is fixed. Sprint deadlines are treated as firm; the team plans buffer time before each milestone rather than extending a sprint.
Quality and technical standards	2	Every feature the team ships must work end-to-end and pass its test cases before being marked done. The team will not knowingly demo or submit a broken feature to protect the schedule.

4.3 Major deliverables and milestones

ID	Milestone or deliverable	Tangible evidence	Target date or sprint	Owner	Reviewer
M-01	Project charter approved	Approved charter document	Sprint 0	Team	Sponsor/Instructor
M-02	Requirements baseline	Use cases, FRs, NFRs, and traceability	Sprint 1	Team	Sponsor/Instructor
M-03	Architecture candidate reviewed	Architecture document and decision record	Sprint 2	Team	Sponsor/Instructor
M-04	Midsemester increment	Working product increment and feedback notes	Sprint 3-4	Team	Sponsor/Instructor
M-05	Feature complete	Testable release candidate	Sprint 6+	Team	Sponsor/Instructor

M-06	Final delivery	Application, tests, documents, and presentation	Final Delivery	Team	Sponsor/Instructor

5 Resources Constraints and Assumptions

5.1 Preliminary resource and budget estimate

Estimate resources at a level appropriate for initiation. Student labor can be recorded in person-hours even when no money changes hands.

Resource	Estimated quantity or effort	Estimated cost	Source or availability	Owner
Student effort	8-10 hours per week	$0 course labor	Team members available course/project time	Project team
Hosting or cloud services	Development and testing using free tier hosting/cloud service	$0	Free tier cloud provider	Development team
Development tools	VS Code, Github, Figma	$0 (free/student licenses)	Free and student developer accounts	Development team
Equipment or test devices	Personal laptops and phones for development and cross-device testing	$0 (existing)	Team members' own devices	Development team
Other	Free-tier AI/LLM API for the AI powered support module (G-03)	$0 (free tier, usage-capped)	Public AI API free tier	Development Team

5.2 Constraints

Constraint ID	Limit placed on the product or project	Source	Effect on the team
C-01	Complete the project within the course schedule	Course calendar	The team must adjust scope before missing a required deadline
C-02	Use only fabricated financial and card data	Sponsor brief	The product cannot connect to real accounts or process real money
C-03	Team members can only commit about 8-10 hours per week due to other coursework	Team capacity	Backlog is prioritized so must-have features (F-01-F-09) are built first;

			nice-to-haves are cut if time runs short)
C-04	AI support module must run on a free or low-cost API tier	Budget ($0 course labor)	Limits response sophistication and request volume; team will add graceful fallback messaging if API limits are hit

5.3 Assumptions and dependencies

ID	Type	Statement	How the team will validate or respond	Owner	Review date
A-01	Assumption	Team members will have sufficient weekly time available to complete development, testing, and documentation	Review progress during regular team meetings and redistribute tasks if necessary	Project team	Each sprint
D-01	Dependency	PostgreSQL is required to store application and user data	Maintain local database backup and use local postgreSQl environment	Project team	Each sprint
A-02	Assumption	Team members have, or can quickly learn the chosen tech stack (frontend, backend, PostgreSQL)	Confirm skills in Sprint 0; schedule short pairing sessions to close any gaps	Project team	Sprint 0-1
D-02	Dependency	The AI-powered support module depends on continued free access to a third-party AI API	Monitor usage against the free-tier limit; design a simple rule-based fallback if the API becomes unavailable	Project team	Each sprint

6 Success Criteria Risks and Agile Working Agreement

6.1 Project success criteria

Criterion ID	What success means	Measure or threshold	Evidence	Reviewer
SC-01	Users can securely register, login, manage their profile, and recover their account	Registration, login, password reset, and profile management pass 100% of acceptance tests	Automated tests, system demo, and acceptance testing report	QA
SC-02	Users can manage budgets, accounts, loans, and financial goals	All required functions work correctly with no defects in testing	Unit tests, integration tests, and test report	QA

SC-03	The application protects sensitive user information	Passwords are secure, sensitive data is not stored in plain text, and security tests pass	Database inspection, security testing, and code review	Backend/QA
SC-04	The AI assistant can properly assist users.	Upon utilizing AI assistant, users can ask for help and receive adequate directions.	Unit tests, integration tests, and test report	QA

6.2 Initial risks and obstacles

Risk ID	Potential problem	Likelihood	Impact	First response	Owner
R-01	Handling sensitive info such as passwords, addresses, and phone numbers creates security and privacy risks	High	High	Minimize stored sensitive information, restrict database access, and perform security testing	Backend
R-02	UI complexity may make application difficult for users with limited financial knowledge	Medium	Medium	Conduct usability reviews and test navigation, readability, and alignment	Frontend
R-03	Incorrect financial calculations could cause users to receive misleading budget, spending loan, or savings info	Medium	High	Define calculation rules early and validate with unit tests and sample financial scenarios	QA
R-04	Free-tier hosting, database or AI API limits affect demo reliability	Low	Medium	Monitor usage against free-tier caps; keep a local backup environment ready for demos	Backend

6.3 Agile working agreement

Working practice	Team agreement
Sprint length and ceremonies	Use 2 week sprints. hold sprint planning at the beginning of each sprint, short checkups during development, and a sprint review at end
Backlog ownership	Team lead maintains backlog. sponsor enters feedback by documentation.
Definition of done	When a feature meets its acceptance criteria, has been code reviewed, has appropriate unit tests, has no defects, and has documentation completed
Communication and response time	The primary channel will be Microsoft teams. Team members should respond within one day unless emergency prohibits them.
Scope change	Team lead approves major change. Approved changes are documented in the project charter.
Conflict or missed commitment	The team first discusses the issue directly and agrees on a solution. If unresolved the issue will be brought up with the professor.

7 Approval and Submission Check

Approval confirms that the reviewers understand the project purpose, boundaries, expected outcomes, and initial authority. It does not freeze the entire backlog. Record approved charter changes in the revision history.

7.1 Review decisions

Reviewer	Role	Decision	Conditions or comments	Date
[Name]	Project sponsor	[Approve / Revise]	[Conditions or comments]	[Date]
Shah, Syed Jawad Hussain	İnstructor	[Approve / Revise]	[Conditions or comments]	[Date]
Jonathan Ochoa	Student project lead	[Accept team commitment]	[Comments]	[Date]

7.2 Signatures

Name and role	Signature	Date
[Project sponsor]		
[Instructor]		
[Student project lead]	Jonathan Ochoa	9/24/26
[Additional approver if required]		

7.3 Final submission checklist

[ ] The problem or opportunity is clear and does not begin with a predetermined solution.

[ ] The charter explains why the project is needed, feasible, and a responsible use of resources.

[ ] Stakeholders, team responsibilities, and decision authority are clear.

[ ] Goals, measurable objectives, and high level features agree with one another.

[ ] In scope and out of scope boundaries can guide backlog decisions.

[ ] Milestones name tangible deliverables, target dates, owners, and reviewers.

[ ] Constraints, assumptions, dependencies, success criteria, and initial risks are recorded.

[ ] The agile working agreement explains how the team will plan, review, test, communicate, and handle changes.

[ ] Every bracketed prompt has been replaced or removed before submission.

[ ] The version, date, author list, and approval decisions are current.

Recommended filename: TeamName_ProjectCharter_v1.0.docx